

★ WEEK 9 · LECTURE

Unsupervised Learning

Grouping, simplifying — and cleaning the noise

Expanded edition · 38 slides · with three hands-on labs

No answer key this week. We find hidden groups (clustering), shrink many features into a few (PCA), and clean the noisy correlations that wreck portfolios.

Where We Are Going Today

Week 8 finished supervised + anomalies. Now: structure without any answer key

1

Finding groups (K-Means)

Sort a messy pile into K natural clusters — no labels needed.

2

Simplifying the picture (PCA)

Squash many features into a few that hold the most information.

3

Cleaning the noise (denoising)

Most stock correlations are random. We separate signal from noise.

By the end you will cluster stocks by hand, extract the market factor, and draw the noise edge.

1

PART 1 · FINDING GROUPS

Finding Groups

K-Means clustering

No Answer Key This Time

Remember the Lego piles from Week 1?

Supervised learning had flashcards with answers on the back. Unsupervised has no answers at all. We just get a pile of data and ask: what natural structure is hiding in here?

Two main jobs

- Grouping: which things belong together? (clustering)
- Simplifying: can we describe it with fewer numbers? (PCA)
- Both find structure, not a labeled answer.

Why finance loves it

- Labels are scarce and expensive in markets.
- Structure is everywhere: sectors, regimes, themes.
- Cleaner structure → better risk models (this week's finale).

Two Worlds, Side by Side

Same course, new rules — what changes when the answer key disappears

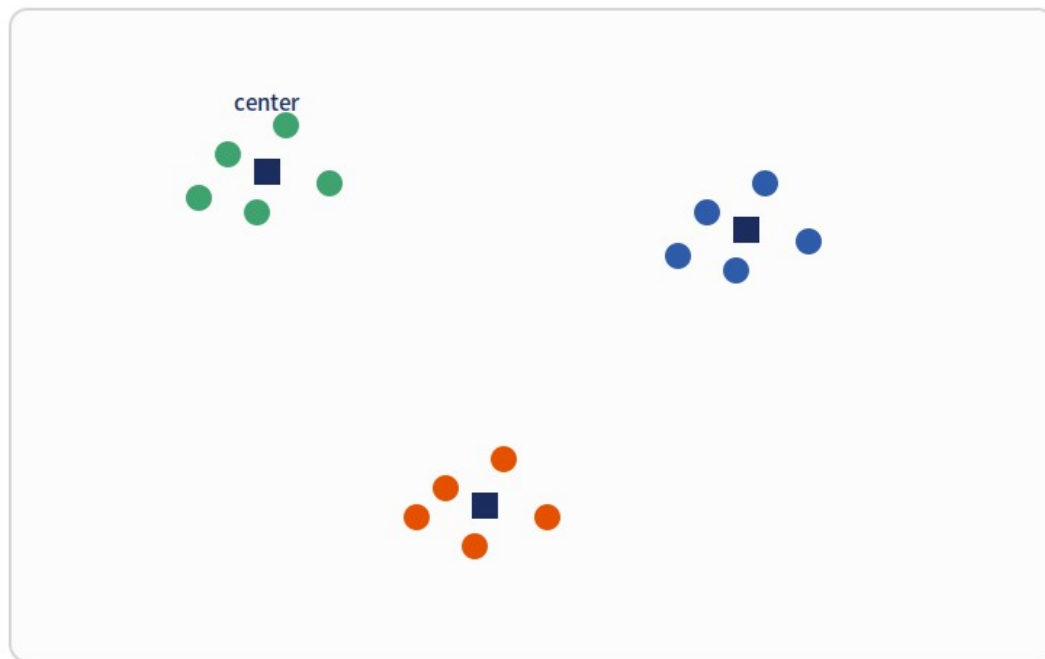
	Supervised (W3–W8)	Unsupervised (today)
Answer key	Yes — labels on every row	None at all
The goal	Predict the label for new rows	Reveal hidden structure
Course examples	Credit default, fraud flag, price direction	Sectors, themes, market regimes
How we judge it	Accuracy, precision/recall, F1 (W3)	Does the structure make sense — and help?

Fine print: our four anomaly detectors were unsupervised all along — they never saw a fraud label.

K-Means — Group by Nearest Center

Pick K centers, assign each point to its closest one, repeat. The groups emerge from the data itself.

K-Means — Sort a Messy Pile into K Groups



The recipe (K = 3 here)

1. Drop K center points at random.
2. Assign each dot to its NEAREST center.
3. Move each center to the middle of its group.
4. Repeat 2-3 until nothing moves.

No answer key needed — the groups emerge from the data itself.

What K-Means Is Trying to Do

Make every point close to its own group's center

$$\min \sum_{\text{points}} \left\| x - \mu_{\text{nearest}} \right\|^2$$

Reading it

- For every point, measure the distance to its nearest center.
- Square it and add them all up.
- K-Means keeps moving centers to make this total as small as possible.

The catch: choosing K

- YOU must pick how many groups K.
- Too few: very different things lumped; too many: groups split for no reason.
- The 'elbow method' helps find a sensible K.

K-Means by Hand — Six Stocks, Two Centers

Cluster by market beta, $K = 2$, starting centers 0.5 and 1.5 (full version: problem set)

Iteration	Cluster 1 (defensive)	Cluster 2 (growth)
1 · assign to nearest	KO 0.5 · PEP 0.6 · JNJ 0.7	AAPL 1.3 · AMZN 1.4 · META 1.5
1 · update centers	$c_1 = (0.5+0.6+0.7)/3 = 0.6$	$c_2 = (1.3+1.4+1.5)/3 = 1.4$
2 · re-assign	unchanged → CONVERGED	unchanged → CONVERGED

Two iterations, done: defensive and growth clusters nobody labeled (real runs restart $n_init=10$ times).

Choosing K: the Elbow

This week's costume for the familiar dial — structure must earn its clusters

K too small

everything lumped

Utilities and meme stocks share one cluster. The total distance J stays large — structure is wasted.

$K = 1$: 'the market is one blob'

K at the elbow

the sweet spot

J drops steeply until real groups are captured, then flattens. Take the bend in the curve.

sweep K , plot J , find the elbow

K too big

noise gets its own club

Groups split on random wiggles; clusters stop meaning anything. Student A, unsupervised edition.

$K = n$: every stock its own cluster

No labels means no F1 — the elbow (plus common sense about the groups) is the validation.

Clustering in Finance

Grouping things that move together

Group similar stocks

- Cluster stocks by how their returns move.
- Discover sectors WITHOUT being told the labels.
- Spot a stock that drifts between groups — a regime change clue.

Group similar customers

- Cluster customers by spending habits.
- Tailor products to each natural segment.
- Powers 'customers like you' recommendations.

Clustering turns a featureless pile into meaningful groups you can act on.

Clustering DAYS, Not Stocks: Market Regimes

Feed K-Means each day's (return, volatility) — calm and storm sort themselves out

The recipe

- One row per DAY: (market return, realized volatility).
- $K = 2$ or 3 typically splits calm / choppy / crisis days.
- No one tells the model what a crisis is — it finds the shape.

Why desks care

- Models trained on calm days fail in storms — switch per regime.
- Position sizing can shrink automatically in the storm cluster.
- A stock drifting clusters = an early warning worth a look.

Same algorithm, new axis: cluster the calendar and the market's moods appear.

2

PART 2 · PCA

Simplifying the Picture

PCA: many features into a few

Why Too Many Features Hurt

The 'curse of dimensionality': in high dimensions data spreads thin and patterns fade. PCA shrinks the room.

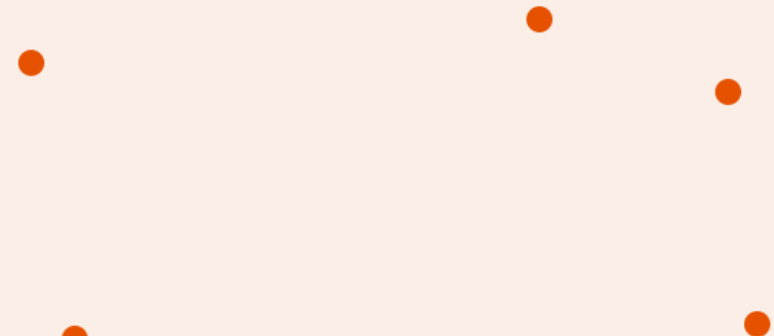
Why Fewer Dimensions Help — Crowded Room vs. Empty Stadium

Few features = a cozy room



points are close → patterns are clear

Too many features = empty stadium



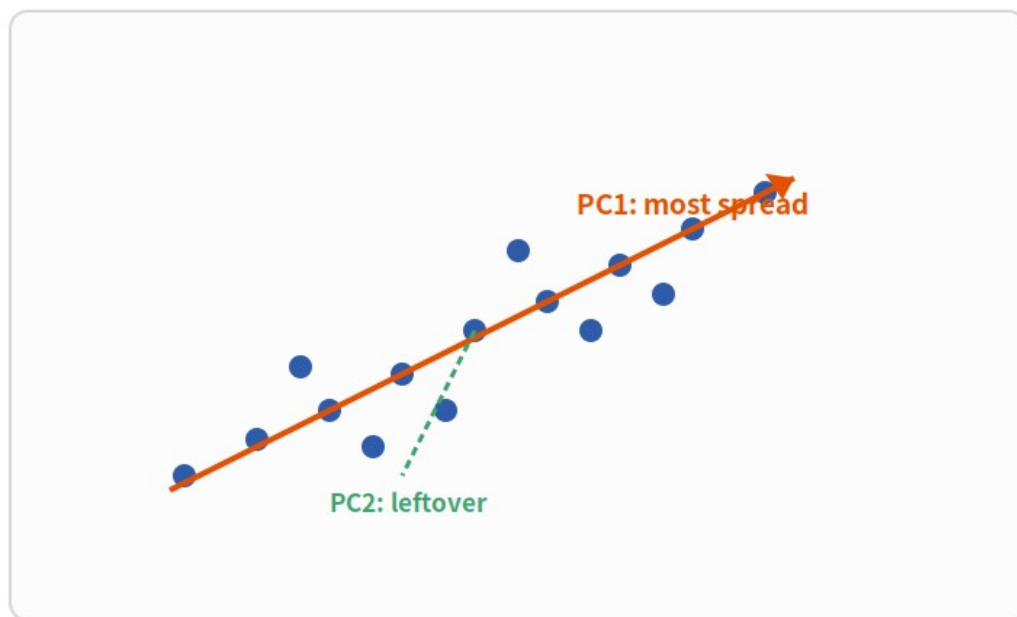
everyone is far apart → patterns vanish in noise

The "curse of dimensionality": more features can mean LESS signal. PCA shrinks the room back down.

PCA — Keep the Most Informative Angle

Find the directions where the data spreads out the most; keep a few, drop the rest — little is lost.

PCA — Find the Most Informative Angle (Like a Shadow)



The shadow idea

A 3D object casts a 2D shadow. From the right angle, the shadow still shows its shape.

PCA finds the angle where the data spreads out the most — that direction holds the most information (PC1).

**Keep a few top directions, drop the rest:
many features become a few, with little loss.**

PCA, Gently

Keep the directions with the biggest spread

$$\Sigma v = \lambda v \quad (\text{keep the few biggest } \lambda)$$

Reading it

- Sigma is the covariance — how features vary together.
- v are special directions (eigenvectors); lambda is the spread along each.
- Big lambda = a direction carrying lots of information.

What we keep

- Sort directions by lambda, biggest first.
- Keep the top few that cover most of the spread; drop the rest.
- Don't compute it by hand in general — one line of code does it.

PCA by Hand — the 2×2 Case You CAN Solve

Two correlated tech stocks: the whole machinery in one clean formula (full version: problem set)

Correlation ρ	Eigenvalues $1 \pm \rho$	Share of spread	PC1 direction	PC2 direction
0.8 (move together)	1.8 and 0.2	90% / 10%	$(1,1)/\sqrt{2}$ — 'both up'	$(1,-1)/\sqrt{2}$ — 'one vs other'
0.2 (barely linked)	1.2 and 0.8	60% / 40%	same directions	same directions

PC1 is the market mode; PC2 is the pairs trade. And the lesson: compression pays only when features are correlated — at $\rho = 0.8$ one number carries 90% of the story.

Reading the Top Components of a Stock Panel

PCA outputs directions; finance gives them names

Component	Typical loading pattern	The name finance gives it
PC1	All stocks, same sign, similar size	THE MARKET — everything up or down together
PC2	Banks + energy vs tech, opposite signs	A sector tilt (value vs growth, rates, oil...)
PC3	Small caps vs large caps	A style factor (size)
PC10 and beyond	No stable pattern, tiny λ	Noise — tomorrow it looks different

The Week 4 factor zoo, rediscovered from raw returns with no labels — and ranked by λ .

PCA in Finance

From 500 stocks to a handful of themes

The returns of 500 stocks are not 500 independent stories. PCA often shows that a few hidden 'themes' explain most of the movement.

What the top components mean

- PC1 is usually 'the whole market goes up/down'.
- Later PCs capture sector or style tilts.
- A few components can explain most daily moves.

Why it's useful

- Cuts the factor zoo (Week 4) down to essentials.
- Speeds up and stabilizes later models.
- Sets up the covariance cleaning we do next.

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PART 3 · CLEANING NOISE

Cleaning the Noise

Denoising the covariance matrix

The Correlations That Lied

Why naive portfolios built on raw correlations blow up

"With 500 stocks but only a few years of data, most of the measured correlations are statistical mirages — pure noise."

The trap

- A covariance matrix needs LOTS of data to estimate well.
- With 500 names there are ~125,000 pairs to estimate — far too many for a few years of history.
- Optimizers then chase that noise and build fragile portfolios.

The fix (López de Prado)

- Use math to find the noise threshold.
- Flatten the noisy part of the matrix; keep only genuine signal.
- A cleaner matrix = a sturdier portfolio.

The Estimation Gap, in Four Numbers

Each correlation is one guess — count the guesses against the evidence

500

Stocks in a typical
institutional universe

~125,000

Correlation pairs to
estimate $(500 \times 499 / 2)$

1,260

Trading days in five
years of history

≈ 0.01

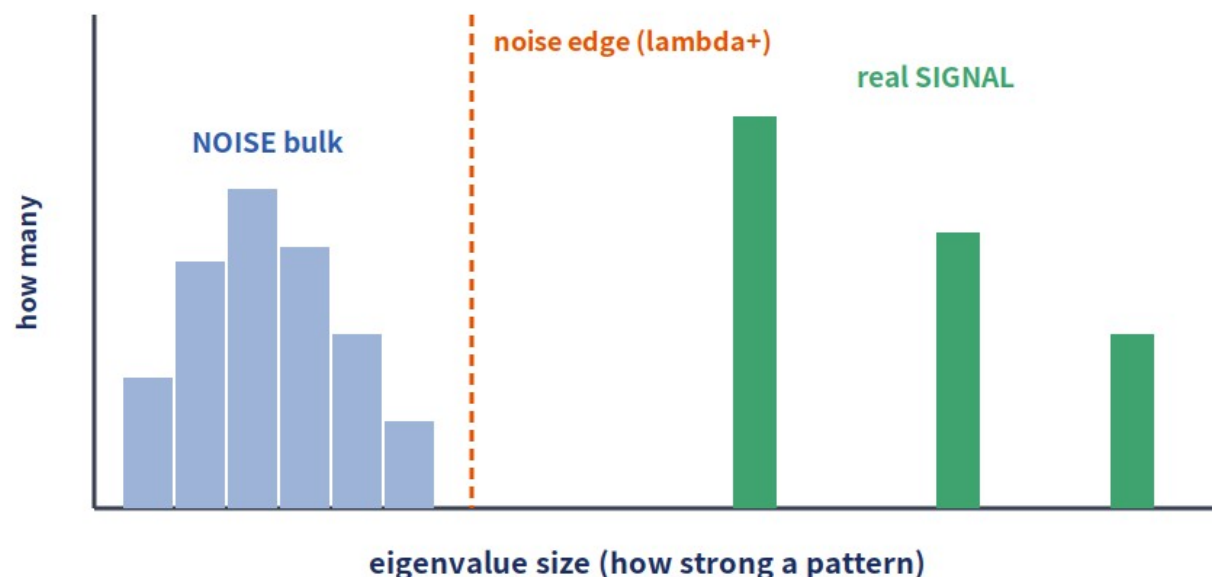
Days of evidence PER
PAIR — a hundred
guesses per fact

One hundred parameters per data point: of course most of the matrix is mirage.

Signal vs. Noise in the Correlations

Marchenko-Pastur draws the line: everything inside the noise bulk is random; only spikes beyond the edge are real.

Most Stock Correlations Are Just Noise



Signal vs. noise

With hundreds of stocks and limited history, most "patterns" are random.

Marchenko-Pastur gives a math line: everything below it is just noise.

Denoise = flatten the noise bulk, keep the few real signal spikes.

A cleaner matrix → better portfolios (Week 11).

The Noise Threshold

N = number of stocks, T = time periods. Eigenvalues below λ_+ are just noise

$$\lambda_+ = \left(1 + \sqrt{N/T}\right)^2$$

Reading it

- More stocks (big N) or less history (small T) → more noise.
- λ_+ is the upper edge of the random 'noise bulk'.
- Anything below it carries no real information.

How we denoise

- Find all eigenvalues of the correlation matrix.
- Replace the ones below λ_+ with their average (flatten them).
- Keep the big signal eigenvalues; rebuild a cleaner matrix.

The Noise Edge by Hand — Two Portfolios

One formula, two very different verdicts (full version: problem set)

Setting	$\sqrt{N/T}$	$\lambda_+ = (1 + \sqrt{N/T})^2$	An eigenvalue of 2.5 is...
Course panel: $N = 10, T = 250$	$\sqrt{0.04} = 0.2$	$1.2^2 = 1.44$	SIGNAL — well above the edge
Big fund: $N = 500, T = 1260$	$\sqrt{0.397} = 0.63$	$1.63^2 = 2.66$	NOISE — below the edge!

Same eigenvalue, opposite verdicts: more names (or less history) push the edge up, and yesterday's 'signal' drowns in the bulk.

Denoising in Three Moves

López de Prado's cleanup, as a checklist

1 · Diagonalize

find the eigenvalues

Compute all N eigenvalues of the correlation matrix — the spread along every direction.

`np.linalg.eigvalsh(corr)`

2 · Flatten the bulk

below λ_+ → the average

Every eigenvalue under the noise edge is replaced by their common average — no fake structure survives.

signal eigenvalues stay untouched

3 · Rebuild

a cleaner matrix

Reassemble the correlation matrix from the fixed eigenvalues and original directions.

feed THIS to the optimizer (W11)

Surgery on the spectrum: keep the organs, remove the mirages.

Why This Sets Up Week 11

Clean inputs are the secret to good portfolios

Portfolio optimizers are extremely sensitive to their inputs. Feed them noisy correlations and they produce wild, fragile, money-losing weights.

Garbage in

- Raw, noisy covariance.
- Optimizer over-trusts random patterns.
- Extreme bets, terrible out-of-sample.

Clean in (Week 11)

- Denoised covariance.
- Methods like HRP and NCO stay stable.
- Sensible, robust portfolios that survive live trading.

The Unsupervised Toolbox, Mapped

Three tools from today — plus four old friends who were unsupervised all along

Tool	Its job	Feed it	Get back	Finance use
K-Means	group similar rows	points	K clusters	sectors, regimes, segments
PCA	compress the columns	returns panel	few components	market factor, themes
MP denoising	clean the spectrum	correlation matrix	sturdier matrix	W11 portfolios
4 detectors (W5–8)	flag the odd one out	unlabeled rows	anomaly scores	fraud screening

None of these ever saw a label — structure, compression, cleaning, and alarms, all from raw data.

Which Unsupervised Tool? — A 60-Second Drill

Cover the right column, answer out loud, then check yourself

Scenario	Which tool?
Do my 10 stocks form natural sector groups?	K-Means on the return series (rows = stocks)
Squash 20 indicators into 3 for a small model	PCA — keep the top- λ directions
Which days were 'crisis mode' last year?	K-Means on daily (return, vol) — regime clusters
Is this eigenvalue of 2.5 real or a mirage?	Compare it to the MP edge $\lambda_+ = (1 + \sqrt{(N/T)})^2$
My optimizer outputs wild extreme weights	Denoise the covariance first — then optimize (W11)

If you got 4 of 5, the toolbox of Week 9 is yours.

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PART 4 · HANDS ON

Your Turn — Hands On

Cluster, compress, and inspect the noise

Getting This Week's Data

Companion notebook — the multi-stock panel (§ 3b)

From the companion notebook (§ 3b)

- Run Setup (§ 0), then the § 3(b) panel loader.
- Daily returns for ~10 stocks in a tidy table `rets`.
- Offline backup (§ 5) gives a simulated panel; same panel as Week 4.

What we'll do

- Lab 1: cluster the stocks and run PCA on returns.
- Lab 2: covariance eigenvalues vs the noise bulk.
- All in scikit-learn + numpy, a few lines each.

[CODE] FML_Data_Setup_Guide.ipynb · § 3 (b) panel loader

```
import yfinance as yf
tickers = ['AAPL', 'MSFT', 'GOOGL', 'AMZN', 'META', 'KO', 'PEP', 'XOM', 'JPM', 'JNJ']
prices = yf.download(tickers, start='2020-01-01', end='2024-12-31',
                    auto_adjust=True, progress=False)['Close'].dropna()
rets = prices.pct_change().dropna() # rows = days, cols = stocks
```

Hands-On Lab 1 — Cluster Stocks & Run PCA

Find groups, then compress the returns

[CODE] colab_lab1.py · data from setup guide § 3b

```
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA
import numpy as np

# cluster stocks by how their returns move (columns)
X = rets.T.values          # one row per stock
labels = KMeans(n_clusters=3, n_init=10).fit_predict(X)
for name, c in zip(rets.columns, labels):
    print(f'{name:6s} -> group {c}')

pca = PCA(n_components=3).fit(rets.values)
print('variance explained:',
      pca.explained_variance_ratio_.round(2))
```

Your task (step by step)

- Load the panel rets (setup guide § 3b).
- Cluster the stocks into 3 groups (K-Means).
- Print which stocks landed together.
- Run PCA; see how much PC1 explains.
- Q: do the clusters match real sectors?

Hands-On Lab 2 — See the Noise in Correlations

Compute eigenvalues and compare to the noise edge

[CODE] colab_lab2.py · data from setup guide § 3b

```
import numpy as np

corr = np.corrcoef(rets.values.T) # stock-by-stock corr
eigvals = np.linalg.eigvalsh(corr) # all eigenvalues

N, T = rets.shape[1], rets.shape[0]
lambda_plus = (1 + np.sqrt(N / T))**2 # noise edge

signal = np.sum(eigvals > lambda_plus)
print('eigenvalues:', eigvals.round(2))
print(f'noise edge = {lambda_plus:.2f}')
print(f'{signal} look like real signal,',
      'the rest are noise')
```

Your task (step by step)

- Build the correlation matrix of rets.
- Compute its eigenvalues.
- Compute the Marchenko-Pastur edge.
- Count how many eigenvalues beat the edge.
- Q: how many are real signal vs noise?

Hands-On Lab 3 — Claude as Your Coding Copilot

AI is allowed in this course — it explains and extends; you understand and verify

[INPUT] Prompt · Claude (paste after your Lab 1 + 2 code)

You are my machine-learning tutor.

- 1) Explain my Lab 1 and Lab 2 code above, one plain-English sentence per line.
- 2) Sweep KMeans K from 1 to 8 on the stock panel; plot inertia vs K and mark the elbow.
- 3) Draw the PCA scree plot (explained variance per component) and print PC1's loadings - do they share a sign, like a market factor should?
- 4) Denoise: flatten all eigenvalues below `lambda_plus` to their average, rebuild the correlation matrix, and print min-variance portfolio weights from the RAW vs the DENOISED matrix side by side.
- 5) In 3 sentences: which weights look saner and why, in this week's garbage-in language.

House rules for AI use

- Allowed: explain, debug, extend code you wrote.
- Required: run it, read it, defend every line.
- Forbidden: submitting code you cannot explain.
- Tip: ask 'why', not just 'do'.

Comprehension Check

Try each question first — the answer is revealed below

Q1 What does K-Means need you to choose in advance?

- A** The labels
- B** The number of groups K ✓
- C** The answer key
- D** The stock prices

Answer: K-Means is unsupervised, but YOU must pick K — how many clusters to form. The elbow helps.

Q2 Why denoise a covariance matrix before building a portfolio?

- A** To make it bigger
- B** Most measured correlations are noise that wrecks optimizers ✓
- C** To add more stocks
- D** To label the data

Answer: With many stocks and little history, most correlations are noise; optimizers chase it and blow up.

Comprehension Check — Round Two

Two more — these cover the new slides

Q3 On a broad stock panel, PC1 almost always turns out to be:

- A** A single lucky stock
- B** The whole-market up/down mode ✓
- C** Pure noise
- D** The smallest eigenvalue

Answer: All loadings share a sign: PC1 is 'everything moves together' — the market factor, found with no labels.

Q4 You add stocks (bigger N), history unchanged. The MP edge λ_+ :

- A** Falls — more data helps
- B** Rises — MORE eigenvalues become 'noise' ✓
- C** Stays fixed
- D** Becomes negative

Answer: $\lambda_+ = (1 + \sqrt{N/T})^2$ grows with N/T : more names on the same history widen the noise bulk — our 1.44 vs 2.66.

Week 9 on One Page

Photograph this slide — it is the whole lecture

Concept	One line to remember
Unsupervised	no answer key — find structure and make it earn its keep.
K-Means	assign to nearest center, move centers to means, repeat; pick K at the elbow.
PCA	keep the biggest- λ directions; PC1 on stocks = the market mode.
2×2 shortcut	corr $\rho \rightarrow$ eigenvalues $1 \pm \rho$; $\rho = 0.8$ means 90% in one direction.
MP edge	$\lambda_+ = (1 + \sqrt{N/T})^2$ — below it, an eigenvalue is indistinguishable from noise.
Denoise recipe	eigenvalues $\rightarrow$ flatten the bulk to its average $\rightarrow$ rebuild $\rightarrow$ then optimize (W11).

Next stop: the graveyard of backtests — why most 'discoveries' die out of sample.

Three Ideas, One Picture Each

The theme: with no answer key, we find structure — and cleaner structure means stronger models

1

K-Means = sort into K groups

Pick K centers, assign each point to its nearest, move centers, repeat. Structure with no labels.

2

PCA = keep the best angle

Find the directions of greatest spread; keep a few, drop the rest. Many features become a few.

3

Denoising = signal vs noise

Most correlations are random. Marchenko-Pastur flags the noise; flatten it, keep the real signal.

Keep the pictures: piles into groups, the best camera angle — and surgery on the spectrum.

Week 9 — Three Things to Remember

Before next class: finish Labs 1–3 and note how few eigenvalues were real signal

1

Unsupervised = structure, no labels

K-Means groups data into K clusters by nearest center. The groups emerge from the data itself.

2

PCA compresses with little loss

Keep the few directions of greatest spread. Beats the curse of dimensionality; reveals hidden themes.

3

Denoise before you optimize

Most correlations are noise. Marchenko-Pastur separates it from signal — vital for Week 11 portfolios.

Next week — Backtesting Pitfalls: purged cross-validation and the Deflated Sharpe Ratio.